

CS 228 : Logic in Computer Science

Krishna. S

Recap

Herbrand's Theorem

Let Γ be a set of equality-free sentences in SNF. Then Γ is satisfiable iff it has a Herbrand model.

Let φ be in Skolem normal form with equality. Then φ is satisfiable iff there is an equisatisfiable formula ψ in Skolem normal form without equality which has a Herbrand model.

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- ▶ $\psi(x_1, \dots, x_n)$ is quantifier-free, and can be written as a conjunction of disjunction of *literals*
- ▶ That is, $\psi(x_1, \dots, x_n)$ can be written in CNF where *literals* are atomic formulae or negations of atomic formulae

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- ▶ Let $H(\varphi)$ be the Herbrand universe of φ
- ▶ Let $E(\varphi) = \{\psi(t_1, \dots, t_n) \mid t_1, \dots, t_n \in H(\varphi)\}$ be the set obtained by substituting terms from $H(\varphi)$ for the variables $x_1, \dots, x_n$ in φ
- ▶ S.T. φ is satisfiable iff $E(\varphi)$ is satisfiable

Herbrand's Method

- ▶ $\varphi = \forall x_1 \dots \forall x_n \psi(x_1, \dots, x_n)$ is satisfiable iff it has a Herbrand model M (by Herbrand's Theorem)
- ▶ φ is satisfiable iff $M \models \psi(x_1, \dots, x_n)$ for all $x_1, \dots, x_n \in H(\varphi)$
- ▶ φ is satisfiable iff $M \models \psi(t_1, \dots, t_n)$ where $t_1, \dots, t_n$ are ground terms
- ▶ φ is satisfiable iff $M \models$ all formulae in $E(\varphi)$
- ▶ φ is satisfiable iff $E(\varphi)$ is satisfiable

Checking Unsatisfiability of φ

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- ▶ $E(\varphi) = \{\varphi_1, \varphi_2, \dots\}$ is a set of quantifier free sentences, so it can be seen as a set of propositional logic formulae
- ▶ Each formula $\varphi_i \in E(\varphi)$ is in CNF
- ▶ We know that $E(\varphi)$ is unsatisfiable iff $\emptyset \in Res^*(E(\varphi))$
- ▶ By Compactness Theorem of propositional logic, there is some finite subset $F = \{\varphi_1, \dots, \varphi_m\} \subseteq E(\varphi)$ such that $\emptyset \in Res^*(F)$
- ▶ So if $\emptyset \in Res^*(\{\varphi_1, \dots, \varphi_m\})$ for some finite m , we conclude φ is unsatisfiable

Checking Satisfiability of φ

- ▶ If $\emptyset \notin \text{Res}^*(\{\varphi_1, \dots, \varphi_m\})$, then we look at $\text{Res}^*(\{\varphi_1, \dots, \varphi_m, \varphi_{m+1}\})$
- ▶ If $\emptyset \notin \text{Res}^*(\{\varphi_1, \dots, \varphi_{m+1}\})$, then we look at $\text{Res}^*(\{\varphi_1, \dots, \varphi_{m+1}, \varphi_{m+2}\})$
- ▶ $\vdots$
- ▶ If φ is satisfiable, then this procedure will continue.

Wrapping Up

- ▶ We have a method to show that a FOL formula φ is unsatisfiable
- ▶ First, write φ in equality free Skolem normal form
- ▶ Check if $\emptyset \in Res^*(E(\varphi))$, this may take some time
- ▶ There is a more systematic resolution for FOL (coming up)
- ▶ We also do not cover a direct undecidability proof for the satisfiability of FOL

Resolution in FOL : A Quick Look

Resolution

- ▶ Let φ be a formula in Skolem Normal Form
- ▶ $\varphi = \forall x_1 \dots \forall x_k \psi(x_1, \dots, x_k)$ with $\varphi_0 = \psi(x_1, \dots, x_k)$ in CNF
- ▶ Let $\mathcal{C}(\varphi_0) = \{C_1, \dots, C_m\}$ be the clauses in φ_0 , C_i is the set of literals in the i th clause
- ▶ $\varphi = \forall x \forall y \forall z ((P(x, y) \vee \neg Q(x, z)) \wedge (R(x, y, z) \vee \neg P(f(x, y), z)))$
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- ▶ $\mathcal{C}(\varphi) = \{ \{P(x, y), \neg Q(x, z)\}, \{R(x, y, z), \neg P(f(x, y), z)\} \}$
- ▶ We identify a formula in SNF with its clauses
- ▶ What is a resolvent of clauses here?

Examples

$$\forall x \forall y \forall z [(\neg Q(x, y) \vee P(f(x), y)) \wedge (\neg P(z, y) \vee R(x, y, z))]$$

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- ▶ Let $C_1 = \{\neg Q(x, y), P(f(x), y)\}$ and $C_2 = \{\neg P(z, y), R(x, y, z)\}$

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- ▶ We have $P(f(x), y) \in C_1, \neg P(z, y) \in C_2$
- ▶ Let $C'_2 = C_2[z/f(x)]$ be obtained by substituting z with $f(x)$

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- ▶ Then $C'_2 = \{\neg P(f(x), y), R(x, y, f(x))\}$ and we can resolve C_1, C'_2

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- ▶ Then $C'_2 = \{\neg P(f(x), y), R(x, y, f(x))\}$ and we can resolve C_1, C'_2
- ▶ C'_2 is a consequence of C_2 : note that x, y, z are quantified for all values
- ▶ $z = f(x)$ is one such : whenever C_2 evaluates to true, so will C'_2

In the following slides, I am using the notation (x/t) to denote substituting variable x with term t . While doing the Translation Lemma, I had used $[t/x]$ to mean the same thing. So please don't get confused, at the end of the day, we only replace a variable with a term.

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- ▶ The resolvent R of C_1, C_2' is a consequence of C_1, C_2' , and C_2' is a consequence of C_2

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- ▶ Thus, R is a consequence of C_1, C_2 also
- ▶ Before finding the resolvent, do appropriate substitutions for variables so that certain literals look the same, **unify** them
- ▶ The substitution $[z/f(x)]$ unified $P(z, y)$ and $P(f(x), y)$

Unification

Let $\mathbb{L} = \{L_1, \dots, L_n\}$ be a set of literals

$\mathbb{L}$ is **unifiable** if there exist variables $x_1, \dots, x_m$ and terms $t_1, \dots, t_m$ such that the substitutions $[x_i/t_i]$ makes all literals $L_1, \dots, L_n$ look the same

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- ▶ Let $sub = (x_1/t_1, \dots, x_m/t_m)$ denote a substitution
- ▶ For a formula φ in SNF, let φsub denote the effect of applying sub to φ .
- ▶ For substitutions sub_1, sub_2 , $\mathbb{L} sub_1 sub_2$ denotes the effect of applying sub_1 followed by sub_2 to all literals in $\mathbb{L}$.

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Also, $sub_1 \geq sub_2, sub_3$. However $\neg(sub_2 \geq sub_3)$ and $\neg(sub_3 \geq sub_2)$.

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- ▶ sub_1 is more general than sub_2 , sub_3 is not a unifier
- ▶ Notations : the first symbol of $P(f(x), y, z)$ is P , second symbol is f , third is x , fourth is y , fifth is z .

Most General Unifier *mg**u*

Let $\mathbb{L} = \{L_1, \dots, L_n\}$ be a set of literals

A substitution *sub* is a *mg**u* for $\mathbb{L}$ if it unifies $\mathbb{L}$, and for any other unifier *sub'*, we have *subsub'* = *sub'*. Essentially, an *mg**u* is at least as general as any other unifier.

- ▶ *sub*₁ is a *mg**u* in the previous example
- ▶ For $\mathbb{L} = \{R(x, y), R(y, x)\}$, *sub*₁ = (x/y) and *sub*₂ = (y/x) are both *mg**u*s
- ▶ $\mathbb{L}sub_1sub_2 = \mathbb{L}sub_2$ and $\mathbb{L}sub_2sub_1 = \mathbb{L}sub_1$
- ▶ *mg**u*s are unique upto renaming

Without a proof

A finite set of literals $\mathbb{L}$ is unifiable iff it has a *mg**u*. We will now see an algorithm for unification.

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- ▶ If the n th symbol of L_i is a variable x , and the n th symbol of L_j is a term t which does not contain x (or vice-versa), then

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- ▶ If any of the hypotheses of the above line does not hold good, then output “ $\mathbb{L}$ is not unifiable”

Example

Let $\mathbb{L} = \{R(f(g(x)), a, x), R(f(g(a)), a, b), R(f(y), a, z)\}$

- ▶ Let $\mathbb{L}_0 = \mathbb{L}$ and $sub_0 = \emptyset$
- ▶ The first and second literals agree till the 3rd symbol, the third literal differs on the 3rd symbol.

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- ▶ The first literal has x as its fourth symbol while the second literal has a (so a term that does not contain x); let $sub_2 = (x/a)$
- ▶ $\mathbb{L}_2 = \mathbb{L}_1 sub_2 = \{R(f(g(a)), a, a), R(f(g(a)), a, b), R(f(g(a)), a, z)\}$
- ▶ $\mathbb{L}_2$ has more than one literal, however now the hypotheses no longer holds, we conclude “ $\mathbb{L}$ is not unifiable”

When is it not unifiable?

- ▶ Discrepancy between literals not having a variable
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- ▶ Discrepancy between a variable and a term containing the variable
- ▶ $P(x, y)$ and $P(f(x), y)$: whatever you substitute for x , the latter will have an extra f

Try Robinson's Algorithm

$$\mathbb{L} = \{f(x_1, g(x_1, x_1), x_2), f(g(y_1, y_1), y_2, g(y_2, y_2))\}$$

- ▶ $sub_1 = (x_1/g(y_1, y_1))$
- ▶ $sub_2 = (y_2/g(g(y_1, y_1), g(y_1, y_1)))$
- ▶ $sub_3 = \dots$

What is the complexity of Robinson's algorithm?

Other unification algorithms : Martelli-Montanari, Escalada-Ghallab

Resolution

- ▶ Let C_1, C_2 be two clauses.
- ▶ Let s_1, s_2 be two substitutions such that $C_1 s_1$ and $C_2 s_2$ have no common variables (note that each clause is Skolemized independently)
- ▶ $\forall x \forall y (C_1 \wedge C_2)$ is $\forall x \forall y C_1 \wedge \forall x \forall y C_2$, so the x and y in C_1, C_2 can be different variables
- ▶ $\forall x \forall y C_1 \wedge \forall u \forall v C_2$

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- ▶ $\forall x \forall y C_1 \wedge \forall u \forall v C_2$
- ▶ Let $L_1, \dots, L_m \in C_1 s_1$ and $L'_1, \dots, L'_n \in C_2 s_2$ be literals such that $\mathbb{L} = \{\neg L_1, \dots, \neg L_m, L'_1, \dots, L'_n\}$ is unifiable.

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- ▶ Let *sub* be a *mg*u for $\mathbb{L}$.
- ▶ Then $R = [C_1 s_1 - \{L_1, \dots, L_m\}] \text{sub} \cup [C_2 s_2 - \{L'_1, \dots, L'_n\}] \text{sub}$ is a resolvent of C_1, C_2 .

Resolution

Let φ be a sentence in Skolem Normal Form, and let

$\mathcal{C}(\varphi) = \{C_1, \dots, C_m\}$ be the clauses of φ

- ▶ Let $Res(\varphi) = \{R \mid R \text{ is a resolvent of some } C_i, C_j\}$
- ▶ $Res^0(\varphi) = \varphi$, $Res^{n+1}(\varphi) = Res(Res^n(\varphi))$
- ▶ $Res^*(\varphi) = \bigcup_{n \geq 0} Res^n(\varphi)$

An Example

Let $\varphi = \forall x(P(a) \wedge (\neg P(x) \vee P(f(x))))$ over a signature containing for constant a , unary function f and unary relation P .

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- ▶ Resolvent $R = [(C_1 - \{P(a)\}) \cup (C_2 - \{\neg P(x)\})]\text{sub}$
- ▶ Resolvent $R = P(f(x))\text{sub} = P(f(a))$

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- ▶ Then we can see that $\mathcal{A} \models P(f(a))$ (point 2 above)
- ▶ $Res^1 = C_1 \cup C_2 \cup \{P(f(a))\}$
- ▶ Now, $\neg P(x) \in C_2 \subset Res^1, P(f(a)) \in Res^1$

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- ▶ Resolvent $R = \{P(f(f(a)))\}$ resolving C_2 and $\{P(f(a))\}$
- ▶ It can be seen that $P(f(f(a)))$ is a consequence of C_2 and $\{P(f(a))\}$
- ▶ $Res^2 = C_1 \cup C_2 \cup \{P(f(a))\} \cup \{P(f(f(a)))\}$
- ▶ $\vdots$
- ▶ $Res^i = C_1 \cup C_2 \cup \bigcup_{i \geq 1} \{P(f^i(a))\}$ and Res^* is infinite

Another Example

$\forall x \forall y \forall z (C_1 \wedge C_2)$ where $C_1 = \{Q(x, y), P(f(x), y)\}$,
 $C_2 = \{R(x, c), \neg P(f(c), x), \neg P(f(y), h(z))\}$

- ▶ $s_1 = (x/u, y/v)$
- ▶ $C_1 s_1 = \{Q(u, v), P(f(u), v)\}$
- ▶ $P(f(u), v) \in C_1 s_1, \neg P(f(c), x), \neg P(f(y), h(z)) \in C_2$
- ▶ $\mathbb{L} = \{P(f(u), v), P(f(c), x), P(f(y), h(z))\}$
- ▶ $\text{sub} = (u/c, y/c, v/h(z), x/h(z))$ unifies $\mathbb{L}$ to $\{P(f(c), h(z))\}$
- ▶ The resolvent is R
 $[(C_1 s_1 - \{P(f(u), v)\}) \cup (C_2 - \{\neg P(f(c), x), \neg P(f(y), h(z))\})] \text{sub}$
- ▶ $\{Q(u, v), R(x, c)\} \text{sub} = \{Q(c, h(z)), R(h(z), c)\}$

Another Example

Is R a consequence of C_1, C_2 ?

- ▶ $C_1 = \forall x \forall y (Q(x, y) \vee P(f(x), y))$ and
 $C_2 = \forall x \forall y \forall z (R(x, c) \vee \neg P(f(c), x) \vee \neg P(f(y), h(z)))$
- ▶ Assume $\mathcal{A} \models C_1 \wedge C_2$
- ▶ $\mathcal{A}$ is over a signature containing functions f, h , relations P, Q, R and constant c
- ▶ Then $\mathcal{A} \models Q(x, y) \vee P(f(x), y)$ for all values of x, y
- ▶ Then $\mathcal{A} \models \forall z (Q(c, h(z)) \vee P(f(c), h(z)))$
- ▶ $\mathcal{A} \models \forall z (R(h(z), c) \vee \neg P(f(c), h(z)))$
- ▶ We can deduce that $\mathcal{A} \models \forall z (Q(c, h(z)) \vee R(h(z), c))$

In general

- ▶ Thus, any resolvent is a consequence of the respective clauses resolved
- ▶ If $\emptyset \in Res^*$, then $\mathcal{A} \models \perp$, and the formula is unsatisfiable.
- ▶ We saw that φ is satisfiable iff $E(\varphi)$ is
- ▶ $E(\varphi)$ is the set of all substitutions of variables in φ with ground terms
- ▶ Whenever we resolve two clauses C'_1, C'_2 from $E(\varphi)$ obtaining a resolvent R' , we also have two clauses C_1, C_2 of φ such that $C'_1 = C_1 \text{sub}_1$ and $C'_2 = C_2 \text{sub}_2$.
- ▶ If R is a resolvent of C_1, C_2 , then there exists some substitution sub such that $R\text{sub} = R'$
- ▶ In other words, any R' obtained using ground resolution in $E(\varphi)$ can also be obtained from φ using first-order resolution.

Lifting Resolution

Lifting Lemma

Let φ be a sentence in Skolem Normal Form. If $R' \in \text{Res}(E(\varphi))$, then there exists $R \in \text{Res}(\varphi)$ such that $R\text{sub} = R'$ for some substitution *sub*.